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Analysis Phase – Polyline Editor Application

1. Introduction:

The purpose of the analysis phase is to understand how users will interact with the Polyline Editor application.This phase focuses on identifying user tasks, workflows, possible issues, and interaction scenarios. (It organize and structure the requirements in detail)

2. User Tasks:

The user can perform the following tasks in the system:

- Start a new polyline
- Add points to create a shape
- Move existing points to adjust the shape
- Delete points from a polyline
- Refresh the screen
- Exit the application

3. Task Flow (Workflow):

The general workflow of the user interaction is as follows:

Start → Press ‘b’ to begin → Click to add points → Press ‘m’ to move points → Drag point → Press ‘d’ to delete → Press ‘r’ to refresh → Press ‘q’ to quit

4. Task Breakdown (Detailed):

Add Point

- User presses ‘b’ to begin
- Clicks on screen
- Point is added and connected to the previous point
- A single point does not form a line, at least two points are required
- The system displays a line only after the second point is added

Move Point

- User presses ‘m’ to activate move mode
- User selects a point by clicking on it
- The selected point is highlighted
- The user drags the point to a new position
- Releases mouse so the previous lines connected to this point are erased and new lines are drawn to it.

Delete Point

- User presses ‘d’
- Selects a point
- Point is removed (Line segments connected to it are erased and other end points of the segments just erased, re-connect with a line segment)

5. User Scenarios:

Scenario 1:-

Jame opens the application and wants to draw a simple house shape. He presses ‘b’ to start a new polyline and clicks on different points to form the structure. He then uses ‘m’ to adjust the corners and ‘d’ to remove an extra point.

Scenario 2:-

Sara creates a polyline but accidentally clicks at the wrong location. She uses the move function to correct the point instead of redrawing the entire shape.

Scenario 3:-

A beginner user opens the application and experiments with different commands to understand how the system works.

6. Problem Identification:

The following usability issues may arise:

- User may select the wrong point accidentally.
- Closely placed points may cause confusion.
- User may forget which keyboard command corresponds to which functionality.
- Deleting a point may distort the shape.
- Lack of visual feedback may confuse users.

7. Suggestions for Improvement:

To improve usability, the following can be considered:

- Highlight selected point
- Show instructions on screen
- Confirm before deleting a point
- Provide visual feedback during actions
- Use simple and consistent controls

8. Clarifications:

Interaction Behavior Explanation

- A single point does not create a visible line; a line appears only after two points are added
- For moving a point, the user must first select it before dragging
- When a connected point is moved, the lines attached to it are automatically updated, giving the effect of line movement.
- If the user clicks on an empty area where no point exists, the system should ignore the action or provide minimal feedback instead of showing intrusive error messages.(pop up on every small mistake = bad UX)

9. Summary:

The analysis phase identifies how users interact with the system, what tasks they perform, and the challenges they may face. This helps in designing a more user-friendly and efficient Polyline Editor application.